THE NOVA RESEARCH CURRICULUM

Mathematics

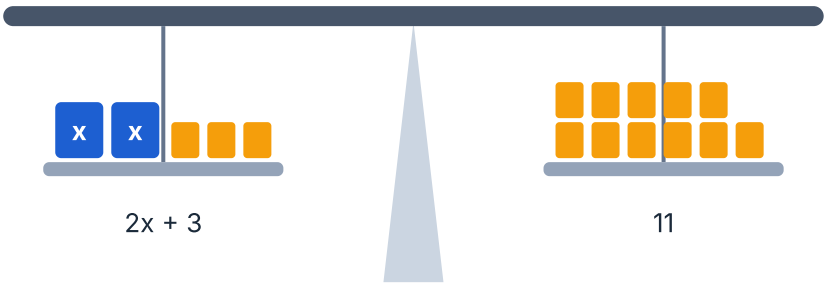
Grade 6 · Term 2

Student edition

Outcomes: MA-6.3 algebraic expressions and linear equations; MA-6.4 lines, angles, constructions and symmetry.

Lesson notes

- **MA-6.3 Expressions (Weeks 1–4):** a letter stands for an unknown number. Terms, coefficients, constants; like terms; substitute values (if x = 3, 2x + 5 = 11).



Each box is x; each small block is 1. Take 3 from both sides, then share by 2: x = 4.

An equation as a balance: 2x + 3 = 11.

- **MA-6.3 Equations (Weeks 5–8):** solve one-step and two-step equations by doing the same to both sides (2x + 3 = 11 → 2x = 8 → x = 4). Check by substituting. Word problems: "Ali's age plus 7 is 19."

  ▶ **Try it on NovaSchool:** Balance-scale equation solver — open **Simulations** on your dashboard.

- **MA-6.4 Geometry (Weeks 9–12):** types of angles; angles on a straight line add to 180°, round a point to 360°; vertically opposite angles are equal; angles in a triangle add to 180°. Construct a 60° angle and a perpendicular bisector with ruler and compass. Lines of symmetry in letters and shapes.
- **Watch out:** adding unlike terms (2x + 3 = 5x); measuring angles from the wrong zero on the protractor.

Workbook pages

Page	Task
1–2	Simplify expressions; substitute values.

Page	Task
3–4	Solve 12 equations; write and solve 4 word problems.
5–6	Find missing angles; constructions with compass.
7	Draw lines of symmetry on 8 shapes.

Question bank

#	Outcome	Question	Your answer
1	MA-6.3	Simplify 3a + 2b + 4a.	7a + 2b / 9ab / 7a + 2 / 9a
2	MA-6.3	If x = 4, find 3x − 5.	12 / 7 / 17 / 2
3	MA-6.3	Solve 2x + 3 = 11.	x = 7 / x = 4 / x = 3 / x = 5.5
4	MA-6.3	Solve x/3 = 6.	x = 2 / x = 9 / x = 18 / x = 3
5	MA-6.3	Ali's age plus 7 is 19. Write and solve an equation.	_____
6	MA-6.4	Two angles on a straight line are 110° and x. Find x.	70° / 250° / 110° / 90°
7	MA-6.4	Two angles of a triangle are 50° and 60°. Find the third.	90° / 70° / 110° / 80°
8	MA-6.4	Vertically opposite angles are always equal. True or false.	True / False
9	MA-6.4	How many lines of symmetry does a square have?	2 / 4 / 1 / 8
10	MA-6.4	An angle of 120° is …	acute / right / obtuse / reflex